Online Appendix

Sourcing under Sanctions: Judicial Urgency and Pharmaceutical Procurement Costs

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Supplementary material for the short paper. All empirical values are generated by the v9 build pipeline and imported through `values.tex`, generated tables, or generated figures.

A. Data Construction and Classifier Validation

The procurement data come from BEC pharmaceutical purchases in group 65 during 2009–2019, linked to litigation and administrative-process markers. The classifier operates at the purchase-order/tender-notice level; the empirical analysis is conducted at the purchase-offer-item (POI) level after classified regimes are linked to BEC item records and sample restrictions are applied. Price regressions use accepted winning bids. The classification rule combines machine-learning predictions, structured JSON flags, regex patterns, and position-based checks, weighted 3, 3, 1, and 1, and classifies a regime when the vote reaches 2; 10 boilerplate patterns are excluded.

Table A.1: Purchase-regime classifier validation.

Validation object	*N*	Agree.	Prec.	Recall	F1	Source
Exact agreement	179,148	98.6%	–	–	–	Ground truth
Judicial class	–	–	0.87	1.00	0.93	Ground truth
Administrative class	–	–	0.92	0.99	0.96	Ground truth
Urgent-class macro-F1	–	–	–	–	0.94	Urgent classes
Judicial class	34,191	–	0.88	0.99	0.93	ML hold-out
Administrative class	34,191	–	0.91	0.98	0.94	ML hold-out

Notes: The classifier operates at the purchase-order/tender-notice level. The table is generated from the production classification report and classifier code. Exact agreement counts 176,713 matched ground-truth labels. The empirical POI-level data link these classified regimes to item-level procurement records.

The classifier universe contains 764,362 classified purchase orders, larger than the 479,330 POI-level analysis file because classification is upstream of item-level linkage. Validation against 179,148 ground-truth purchase orders gives 98.6% exact agreement, with urgent-class F1 of 0.93 (judicial) and 0.96 (administrative) and macro-F1 0.94. A purchase-order-level error-sensitivity diagnostic (not tabulated) implies contamination rates that are small relative to the regime differences studied here; remaining misclassification would attenuate, not inflate, the regime contrasts.

Table A.2: Sample construction and empirical units.

Object or sample	Size	Unit and use
Classifier universe	764,362	Purchase orders/tender notices classified into procurement regimes.
Full BEC pharmaceutical file	479,330	Purchase-offer-item observations after linking BEC item records to regime classifications.
Analysis sample	226,305	POI observations satisfying core item and variable restrictions.
Winners-only price sample	196,883	Accepted winning bids used in negotiated-price regressions.
Urgent panel	56,803	Administrative and litigated urgent winning bids used for the under-the-gun comparison.
Both-regime urgent cells	5,581	Item-by-year-month cells containing both administrative and litigated urgent purchases.
Singleton urgent cells	11,972	Comparable urgent cells containing only one urgent regime.
Firm-buyer-item triples	1,206 triples; 4,573 observations	Same supplier, buyer, and item comparisons used to isolate within-firm pricing.

Notes: This table separates the classifier unit from the empirical regression units. POI denotes purchase-offer-item. Price regressions use accepted winning bids, while classifier validation operates at the purchase-order/tender-notice level.

The analysis sample contains 226,305 POI observations; the winners-only price sample 196,883 accepted winning bids; the urgent panel 56,803 winner observations; and the firm-buyer-item triple sample 4,573 observations in 1,206 triples. Administrative and litigated urgent purchases are not balanced by design: administrative purchases are selected, larger, and differ in item composition. The design therefore combines selection bounds, within-firm pricing tests, and direct sourcing evidence rather than treating the administrative channel as randomly assigned.

B. Selection Bounds

Administrative requests are screened before purchase, which likely makes administrative cases easier or cheaper to fulfill, so naive administrative-versus-litigated comparisons can overstate the sanction-related gap. The main paper's Table 1, Panel B reports the resulting Lee bounds; this section documents the trimming and its robustness. All coefficients are administrative minus litigated log negotiated prices, so a negative coefficient means litigated purchases are more expensive;

percentage gaps are reported in the reader-facing litigated-over-administrative direction. Trimming the high and low tails of the administrative price distribution within item × year × PBU strata gives the lower and upper bounds. Trimming affects 12,038 of 44,654 strata, with mean rate 26.9% and maximum 100.0%. A parametric (Heckman-type) selection correction is non-informative in this design—the available exclusion is weak and the fixed effects absorb most identifying variation—so the bounds plus mechanism evidence are preferred to a point estimate.

Table B.1: Lee bounds under alternative trimming strata.

Strata	Admin coef.	Gap (%)	Trim (%)
item × year	[-0.326, -0.202]	[22.4, 38.6]	23.1
item × year × PBU	[-0.192, -0.148]	[15.9, 21.1]	26.9
item × year-month × PBU	[-0.202, -0.202]	[22.4, 22.4]	26.8

Notes: The outcome and second-stage fixed effects are held fixed across rows: log negotiated price with item, year, and PBU fixed effects. Rows vary only the strata used to compute the administrative trimming share. The item × year × PBU row is the preferred specification in the main analysis. Coefficients are administrative minus litigated log prices; percentage gaps are reported as litigated-over-administrative prices. In the finest item × year-month × PBU strata, cells contain too few administrative observations for top-tail and bottom-tail trimming to remove different observations, so the lower and upper bounds coincide; this row is reported for completeness, and the preferred bounds use the coarser item × year × PBU strata.

C. Robustness and Diagnostics

Placebo. The placebo uses items never observed in litigation, checking whether the urgent-procurement pattern appears outside the litigation margin rather than reflecting generic platform or time-trend dynamics.

Inference. Because PBU clusters are few and uneven, Rademacher wild-cluster inference for the under-the-gun contrast gives $p = 0.0080$ for the preferred specification and $p = 0.0390$ for the tighter item-by-year-month specification; the contrast continues to reject a zero gap, though inference does not address the selection concern handled by the bounds.

Channel diagnostics. Adding supplier fixed effects and log quantity attenuates the urgent price coefficient, consistent with sourcing and scale rather than a broad same-firm markup. Conditioning tightly on log quantity is a channel diagnostic, not a robustness test of the total effect: quantity is itself a margin of judicial urgency, so the coefficient can change sign once the scale channel is partialled out, and the preferred total object remains the no-quantity-control bounded estimate.

Table C.1: Placebo Test on Never-Litigated Items

	Negotiated price		Reference price	
	Never-litigated	Main sample	Never-litigated	Main sample
Urgent purchase	-0.020	0.051***	-0.031	0.031**
	(0.032)	(0.015)	(0.045)	(0.014)
Observations	39,283	196,883	46,874	226,278
R^2	0.834	0.868	0.811	0.852
Item FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
PBU FE	Yes	Yes	Yes	Yes

Notes: The never-litigated sample contains items with zero litigated purchases during the sample period and variation between ordinary and administrative urgent purchases. The main sample contains items observed in both ordinary and litigated procurement. All specifications include item, year, and PBU fixed effects. Standard errors, in parentheses, are clustered by PBU. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A direct aggregation comparison within common buyer-item-month cells is mixed—administrative cells carry larger total quantity while litigated cells show more repeated purchase-offer-items—and is suggestive of fragmented repeated demand only in combination with the main-paper decomposition figure and winner-switching evidence. Heterogeneity of the within firm-buyer-item test across quantity, formulary, and period subsamples is reported in the main paper's Table 2; the deep-market null and the reappearance of supplier leverage in thinner or earlier subsamples are both part of the mechanism evidence.

Dynamics. The BJS event study assesses timing around item-level exposure and is diagnostic, not the main identification strategy. The first post-exposure estimate is 0.052 (SE 0.018) and the five-period estimate is 0.147 (SE 0.026). Honest-DiD sensitivity does not survive deviations at the observed maximum pre-period scale, so the main claims rest on the selection bounds, the within firm-buyer-item pricing test, and the sourcing evidence.

Table C.2: Dynamic-design sensitivity summary.

Diagnostic quantity	Value	Interpretation
BJS coefficient at first post period	0.052	First post-exposure timing estimate
Standard error	0.018	BJS period-specific standard error
Maximum absolute pre-period coefficient	0.041	Observed pre-period deviation scale
Breakdown linear-extrapolation M	0.018	Smallest linear-extrapolation allowance that reaches zero
Survives observed pre-period maximum	no	Diagnostic robustness indicator

Notes: The BJS event study is used to assess timing patterns, not as the primary identifying design. The final row records whether the first post-period estimate remains different from zero when the allowed post-period violation equals the observed maximum absolute pre-period coefficient. The sensitivity calculation uses the saved BJS period-specific standard errors; the off-diagonal covariance is unavailable in the source BJS event-study output.

D. Procurement-Cost Calculation

The fiscal procurement-cost calculation applies the Lee midpoint 18.5% to an admissibility calibration of 50.0% and annual litigated spending of \$300 M, yielding \$27.8 M per year (Lee-bound range \$23.9 M–\$31.7 M). This is a fiscal procurement-cost calculation, not a full welfare estimate: it excludes health benefits, search and compliance costs, and other welfare components, and it should not be read as the value of scaling back legal enforcement. Varying the admissible share of litigated spending or the annual-spending anchor moves the figure mechanically and leaves the bounded under-the-gun gap unchanged.

Table D.1: Fiscal procurement-cost implication of the bounded UTG gap.

Component	Value
Lee midpoint	18.5%
Lee lower bound	15.9%
Lee upper bound	21.1%
Admissibility calibration	50.0%
Annual litigated spending	\$300 M
Annual procurement-cost implication, midpoint	\$27.8 M
Annual procurement-cost implication, lower	\$23.9 M
Annual procurement-cost implication, upper	\$31.7 M

Notes: This calculation is not a full welfare estimate. It applies the bounded price gap to a calibrated admissible share of litigated spending and excludes patient health benefits, search costs, and other welfare components. It should be read together with the main-paper sourcing decomposition.